POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2018
Programme: BE | Full Marks: 100
Course: Artificial Intelligence | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is Artificial Intelligence? Mention the advantages of Artificial Intelligence. 8
   b) What are the criteria for the evaluation of search algorithm? Compare DFS and BFS. 7
2. a) Solve the following problem : 8
   SEND+MORE=MONEY
   b) Write algorithm for A*. How we can optimize the A* algorithm? 7
3. a) Represent following statements into predicate logic. 8
      i) All Hindu are either loyal to Krishna or Shiva.
      ii) Every gardener like sun.
      iii) There is exactly two red mushrooms.
      iv) Every parents are older than their childs.
   b) Define Bayes rule for probabilistic problem. If probability of symptoms on patients is 1/20 and probability of disease known by doctor is 1/45000. Also the probability of symptoms condition to disease is 50%. Now, calculate the probability of disease condition to symptoms using Bayes theorem. 7

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4. a) Define deep learning. Explain genetic algorithm. 7
   b) Define PCA. Compare Brain and computer in terms of intelligence, memory etc. 8
5. a) What is an expert system? Explain about its knowledge. 7
   b) Solve the AND function using Hebb network. 8
6. a) Why natural language understanding is difficult? Justify your answer. 7
   b) What do you mean by machine learning? Explain learning by anology. 8

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7. Write short notes on: **(Any two)**

a) Kohonen Network

b) Turing Test

c) Informed search Vs Uninformed search

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